AI workflow automation with n8n

BC3415 Week 10 · Jonathan Low · n8n 2.44.6 self-hosted (local) · Telegram + Google Gemini · synthetic data only

1. Summary

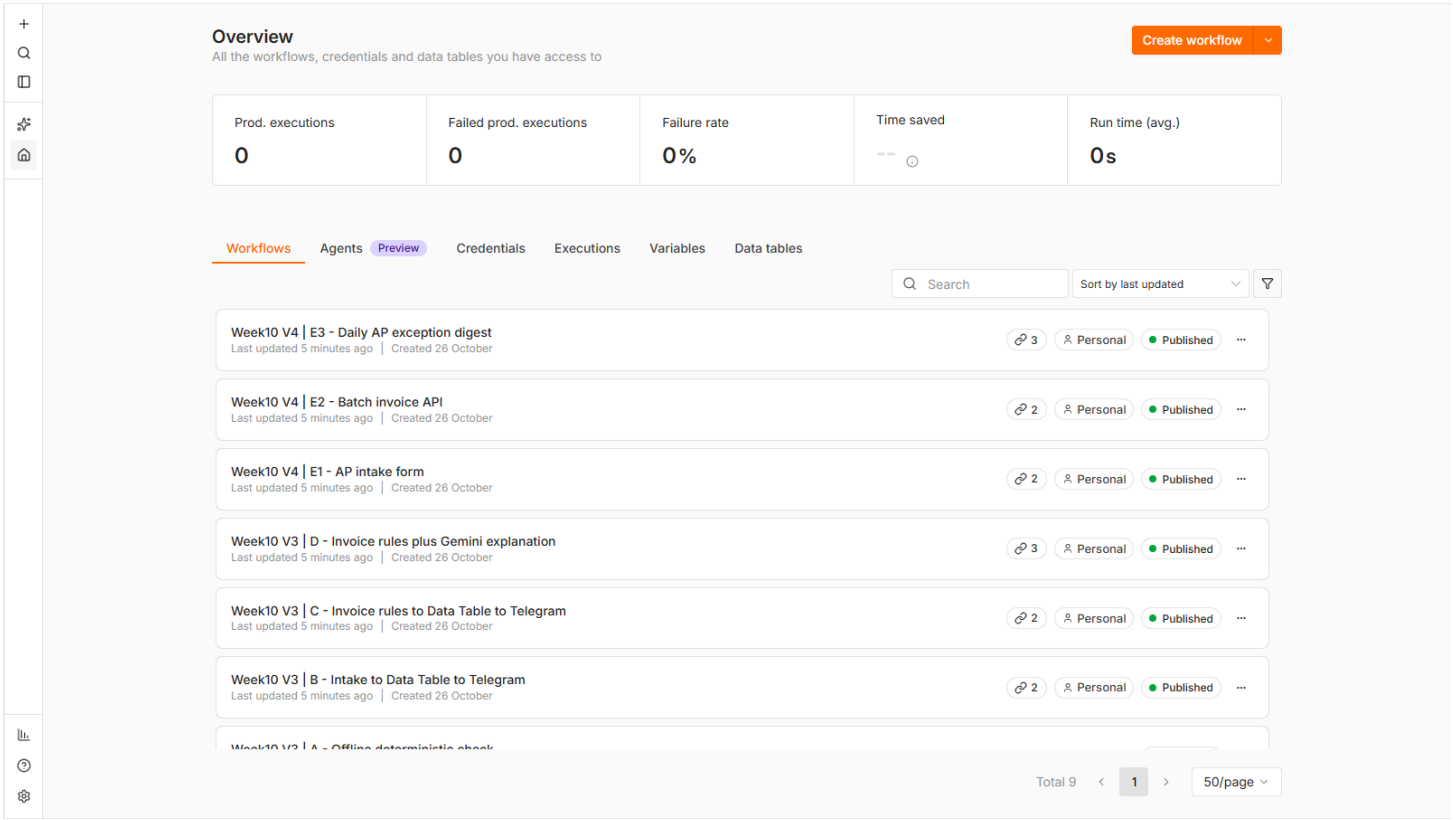
I installed and ran n8n locally, imported and configured all four lab workflows (A–D) with my own Telegram bot and Gemini key, and ran the guide's test cases. Every checkpoint on the lab's evidence sheet is met, and every execution is logged in `execution_logs/`. I then built a customised **V4 accounts-payable triage service**: five workflows that fix the limitations the lab guide itself lists (Appendix D and section 7.4): a hard-coded PO list, a duplicate check that never learns, and AI text that is never stored. V4 also adds new fraud controls, priority-based alerting, a batch API, a guardrail on the AI output, and a daily manager digest.

Workflows built and run	Executions logged	Telegram messages sent	Unit tests passing
9	**26**	**16**	**22**

Payments authorised
0

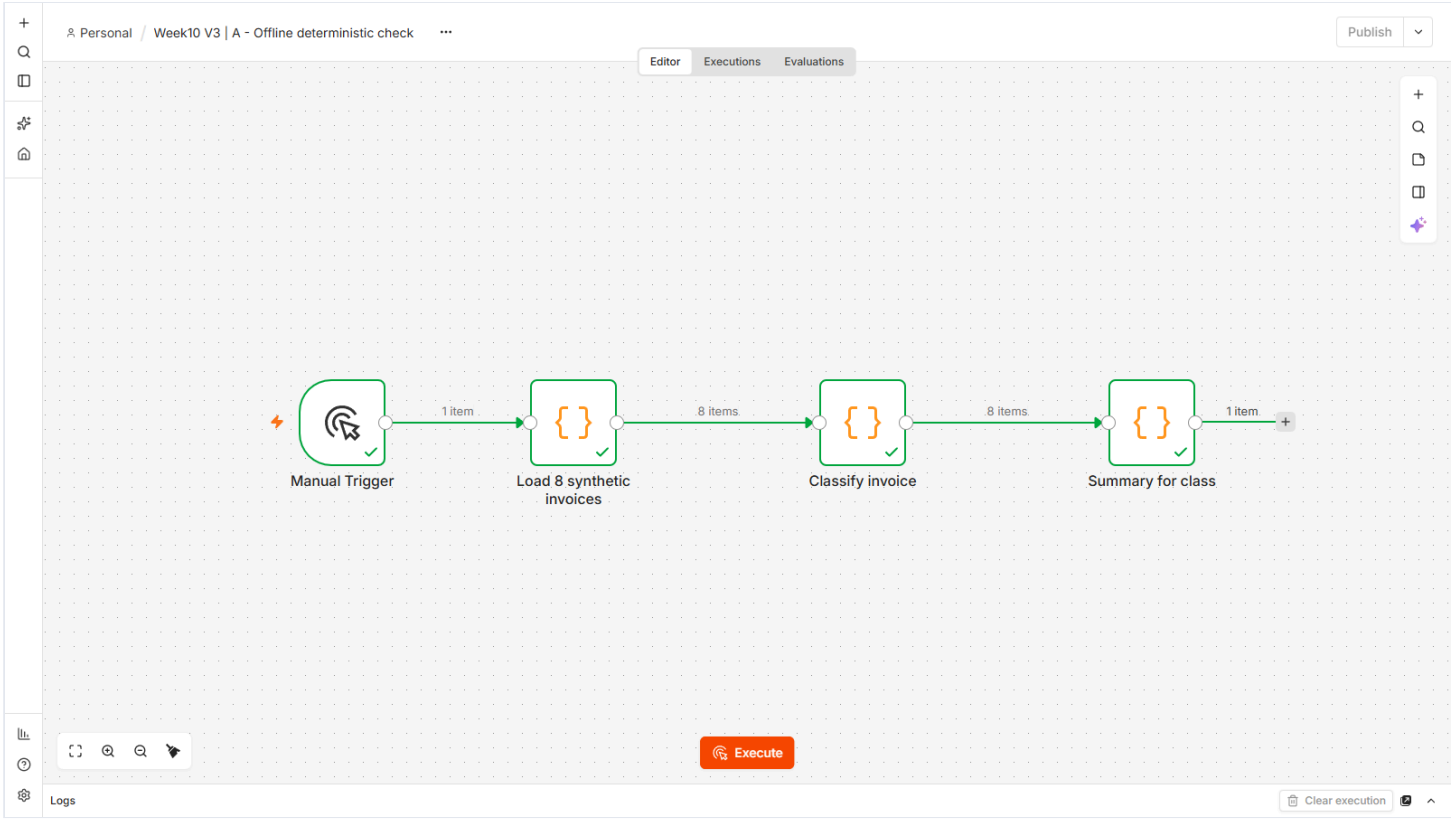
2. Setup

- **Route A (local)**, without Docker: n8n 2.44.6 installed with npm into a project folder, using a portable Node 24 (n8n 2.x needs Node ≥ 24; the machine's global Node 22 was left alone). Editor at `http://localhost:5678`, telemetry off, timezone Asia/Singapore.
- Telegram: my own bot from @BotFather. The recipient is my numeric private chat id (from `getUpdates` after sending `/start`), not the bot's id, which avoids the guide's "bot can't send messages to the bot" 403 error.
- Gemini: Google AI Studio key, stored as an n8n credential.
- **No secrets in this submission.** Keys live only in n8n's encrypted credential store. The chat id is read at run time as `{{ $env.TELEGRAM_CHAT_ID }}`, so it isn't in any workflow JSON. Chat id, name and username are masked in screenshots and logs.
- Reproducible: `tools/n8n_setup.py` rebuilds everything on a fresh n8n: Data Tables, seeded PO master, credentials (from environment variables), workflow import with id wiring, and publishing.

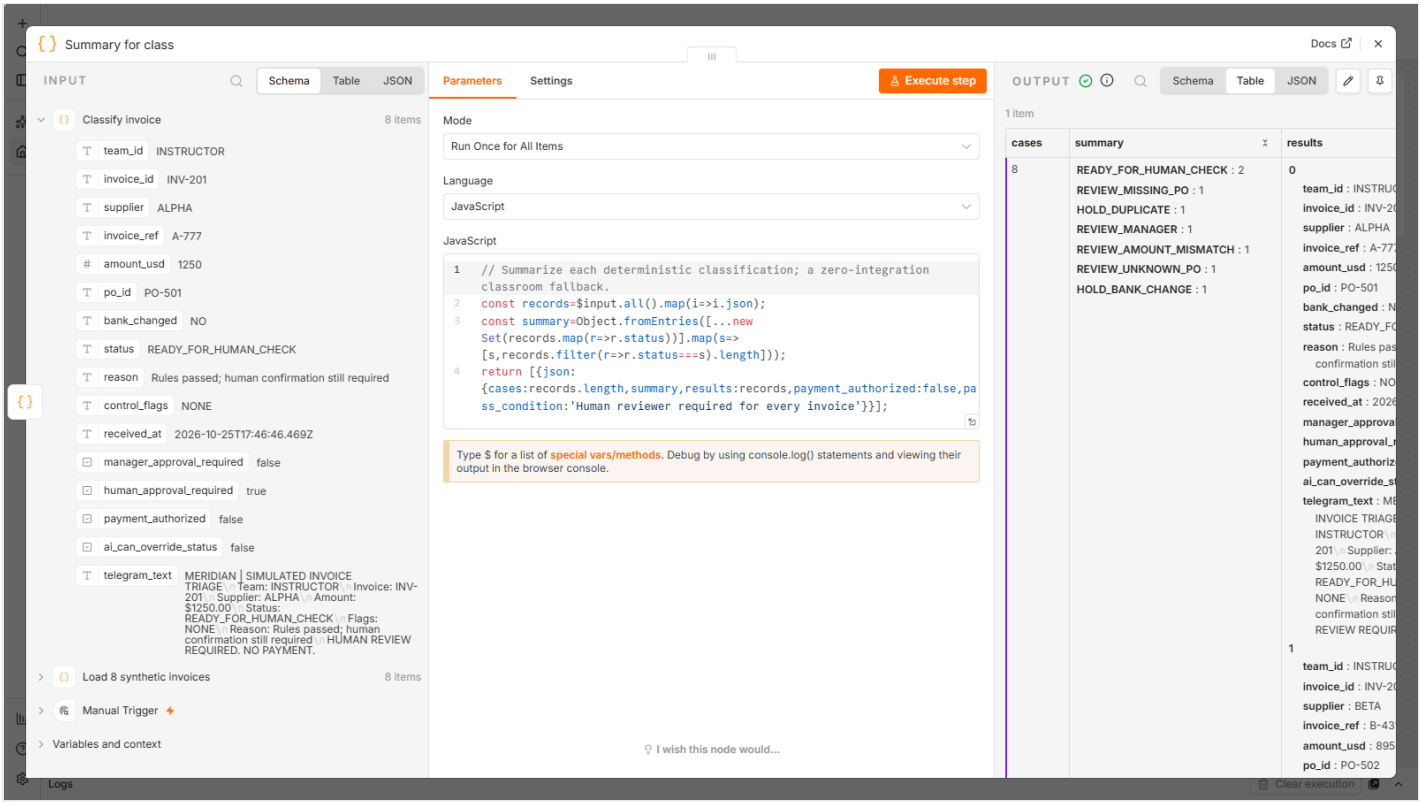


All nine workflows imported and published.

3. Lab workflows A–D (as supplied, configured)

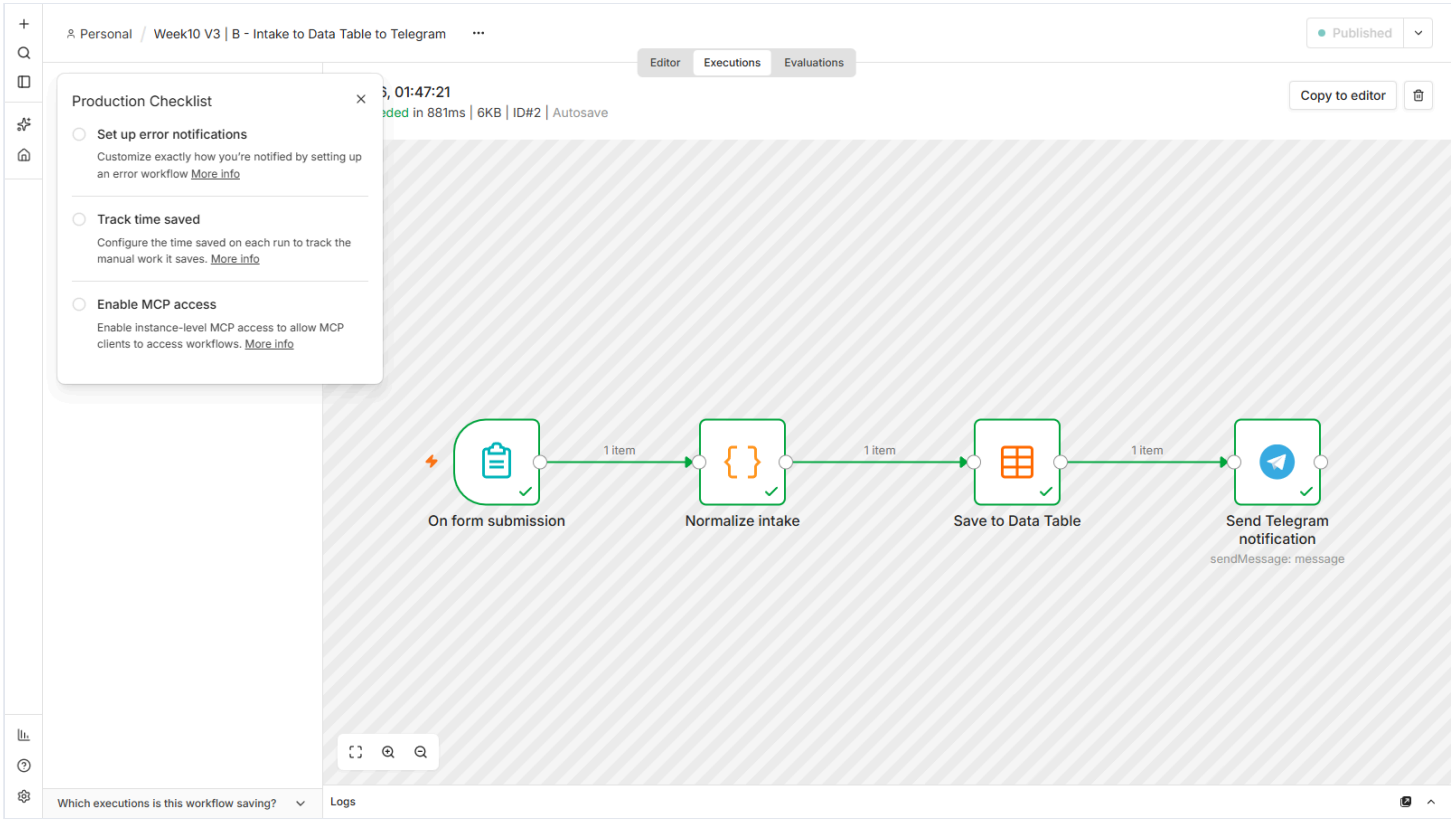


A: offline check, 1 → 8 → 8 → 1 items.

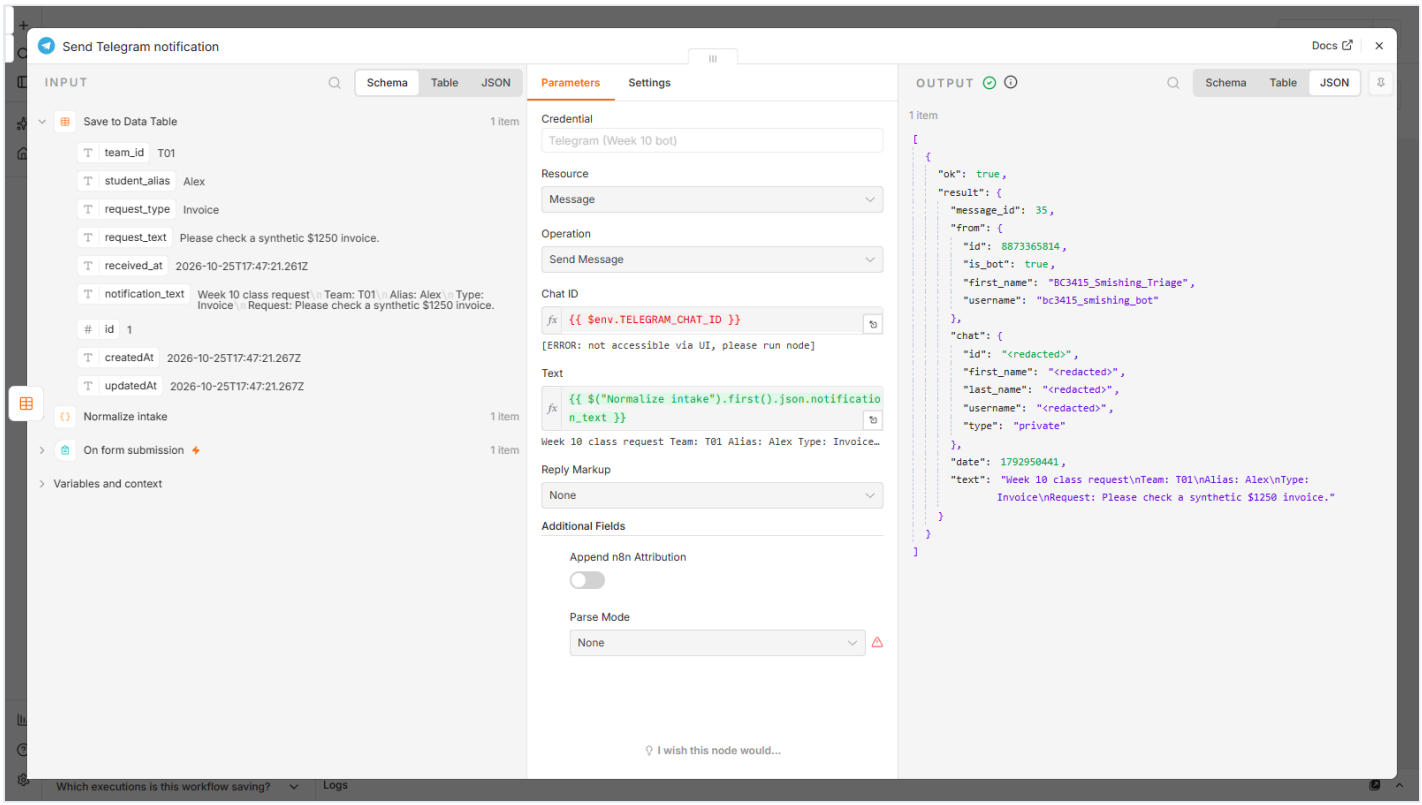


A: Summary for class: cases = 8, two READY and one of each other status, exactly the guide's Appendix A answer.

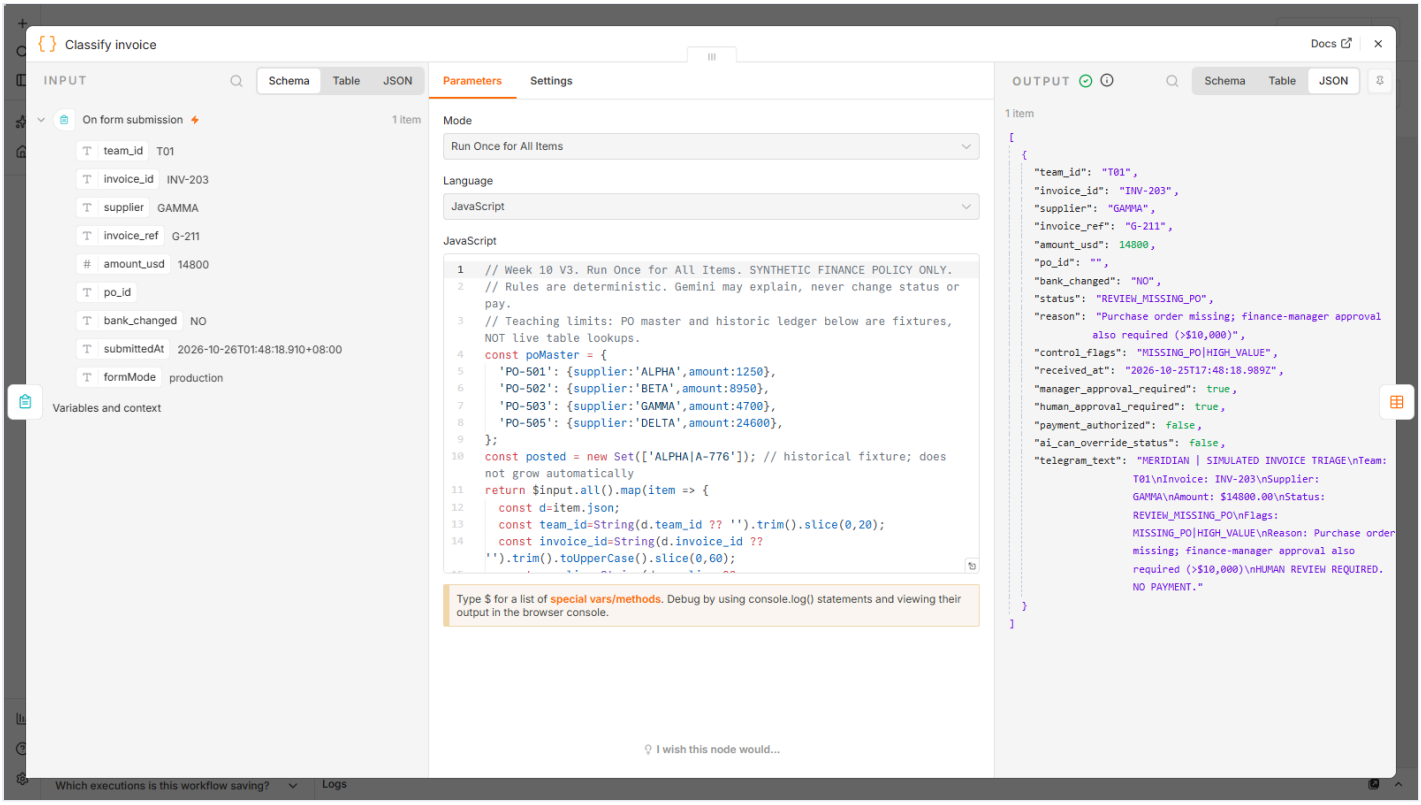
Flow	Case	Execution	Result	What happened	Telegram
B	INV request T01 / Alex	#2	success	row saved to Week10_Intake	ok=true, message_id 35
C	INV-201	#3	success	READY_FOR_HUMAN_CHECK (NONE); manager=false, paid=false	ok=true, message_id 36
C	INV-203	#4	success	REVIEW_MISSING_PO (MISSING_PO|HIGH_VALUE); manager=true, paid=false	ok=true, message_id 37
C	INV-204	#5	success	HOLD_DUPLICATE (HISTORIC_DUPLICATE); manager=false, paid=false	ok=true, message_id 38
C	INV-205	#6	success	REVIEW_MANAGER (HIGH_VALUE); manager=true, paid=false	ok=true, message_id 39
D	INV-203 (team T02)	#7	error	Gemini 3.8 Flash: 503 Service unavailable (see 6.1)	not sent
D	INV-203 (team T02)	#8	success	REVIEW_MISSING_PO + AI note	ok=true, message_id 40



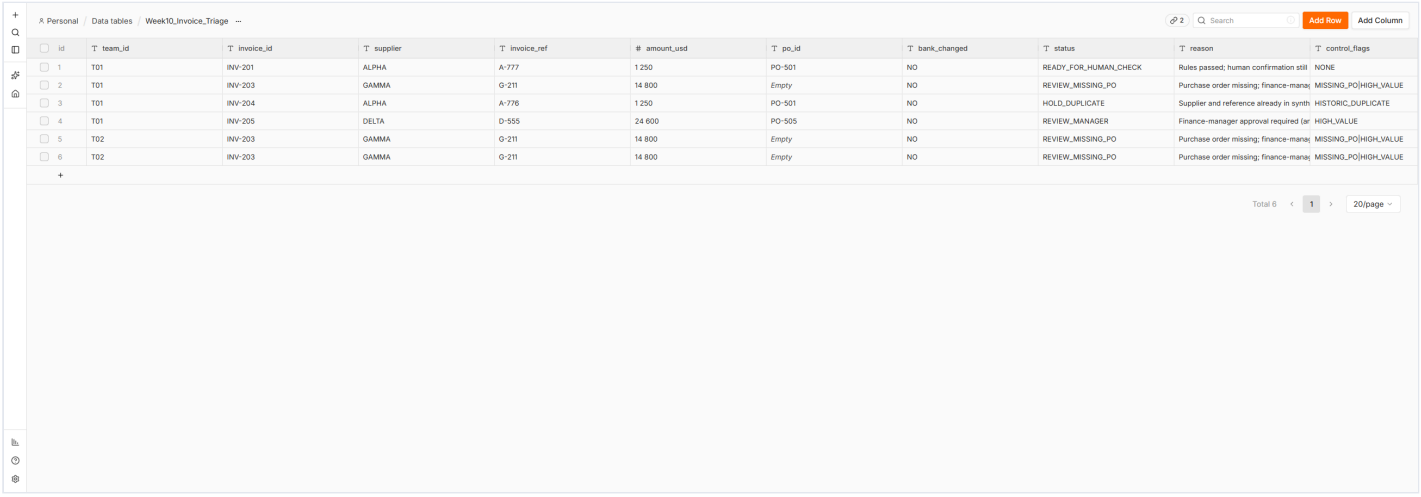
B: form → Normalize intake → Data Table → Telegram, all green.



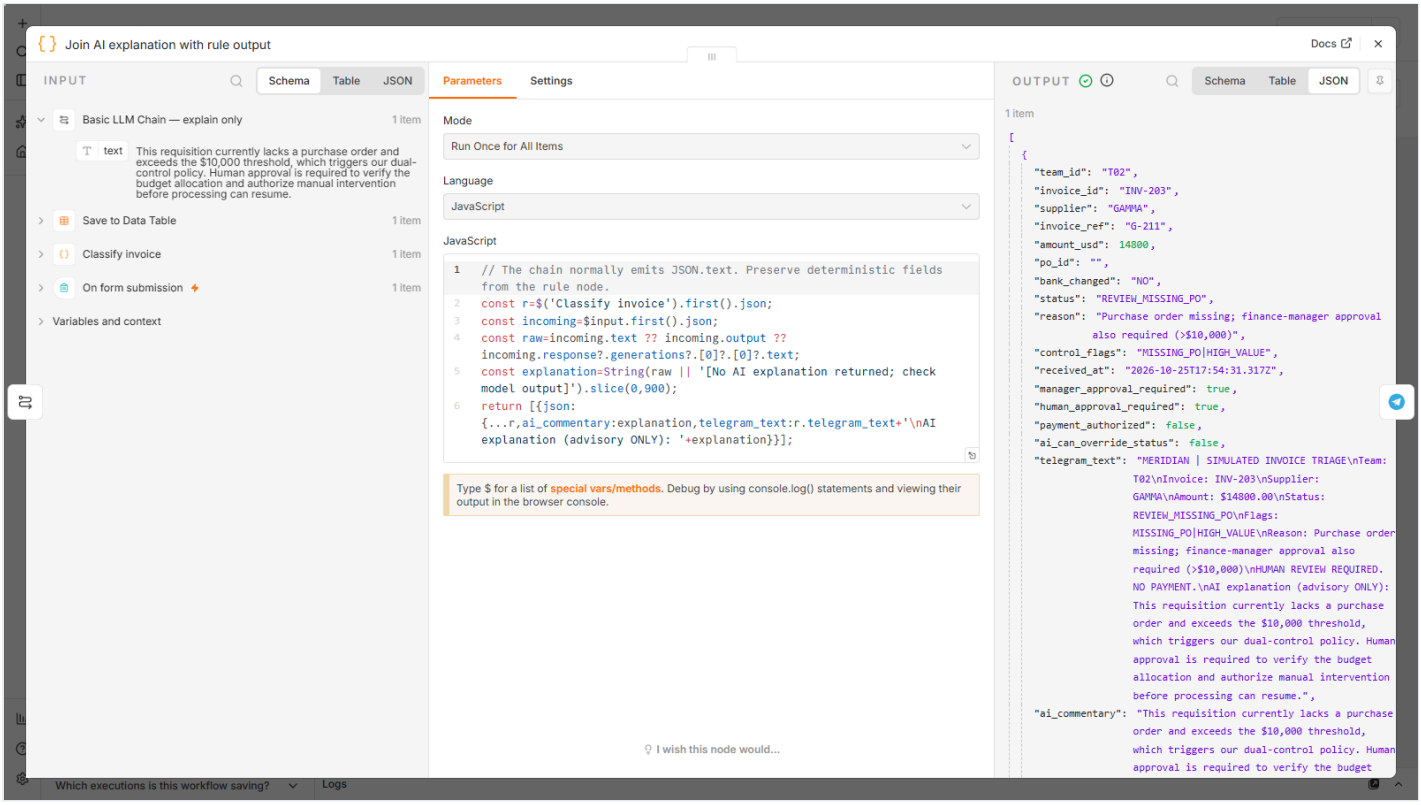
B2 checkpoint: Telegram API returned ok=true with a message_id (recipient details masked).



C2 checkpoint: INV-203 is REVIEW_MISSING_PO with MISSING_PO|HIGH_VALUE, manager_approval_required=true, payment_authorized=false.



C1 checkpoint: Week10_Invoice_Triage with its 16 application columns and the C/D rows.



D: Gemini's explanation joined to the unchanged rule output; payment_authorized stays false.

Evidence sheet (guide section 9.1): A, B1 (`T1_datatable_Week10_Intake.png`), B2, C1, C2, C3 (INV-204 HOLD_DUPLICATE and INV-205 REVIEW_MANAGER in the table above), and D are all met, with execution logs for each.

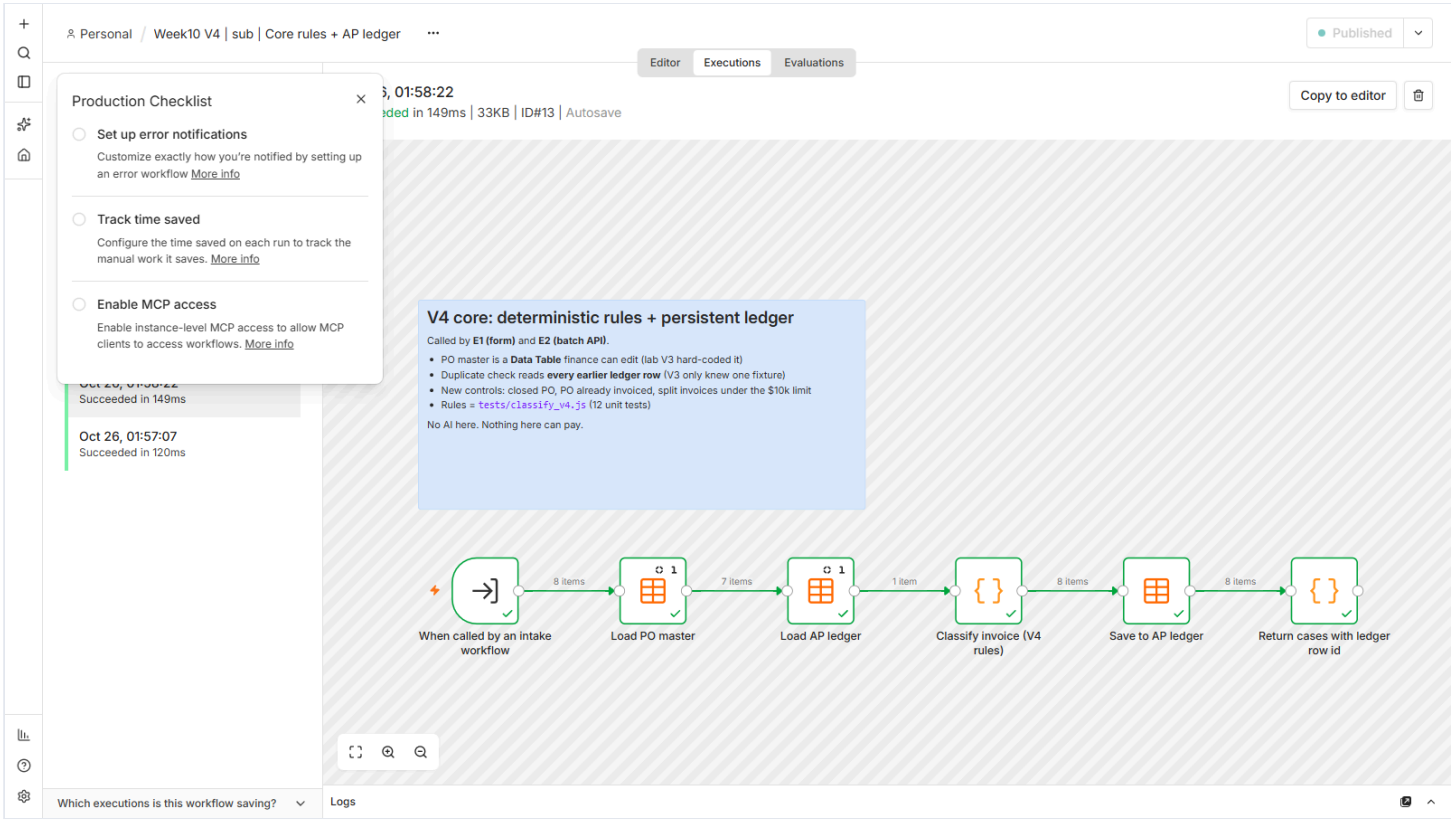
4. Customisation: V4 AP triage service

4.1 Business problem

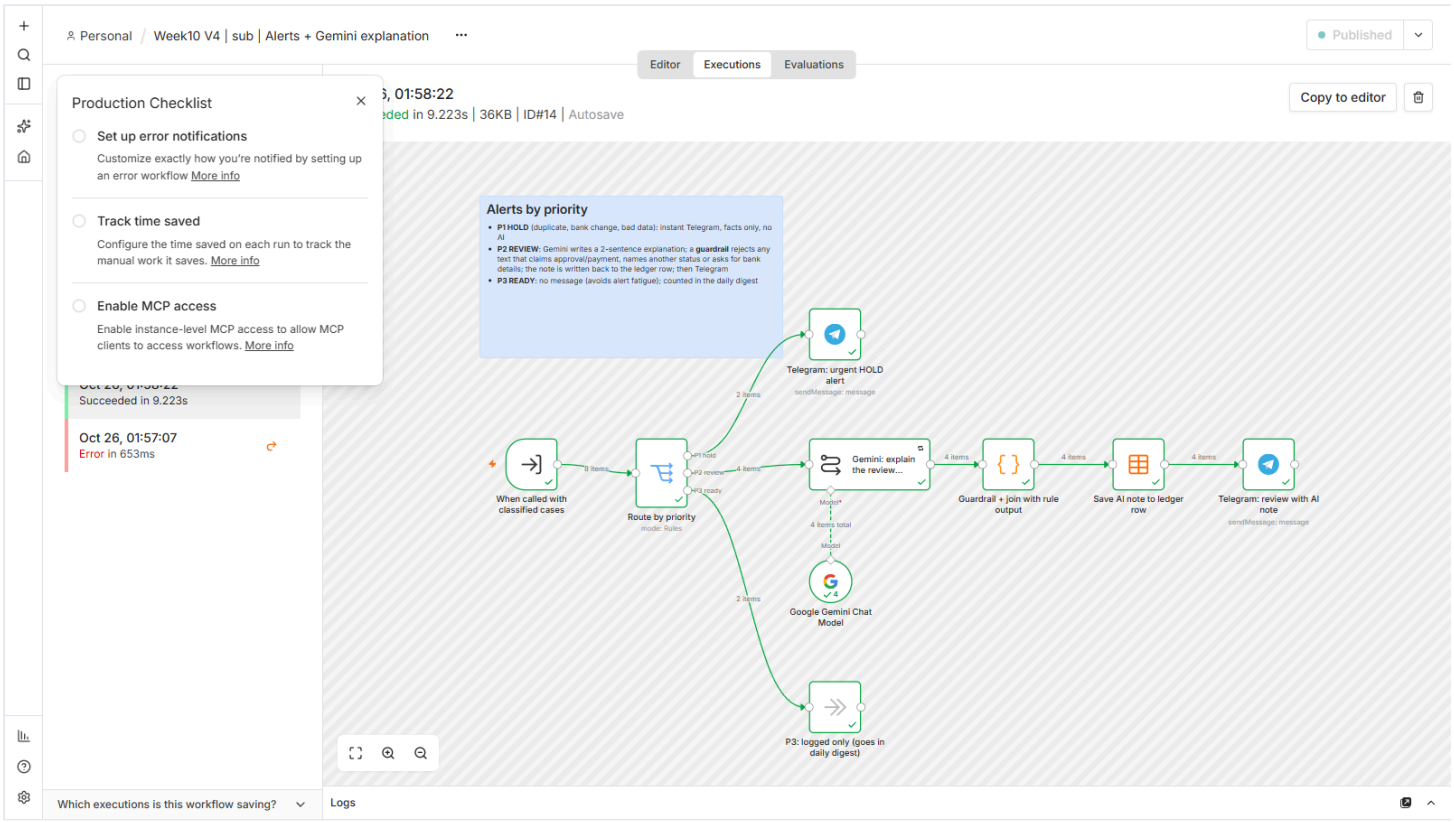
Workflow C is a good teaching template, but the guide lists why it can't be used for real: the PO list is fixed in JavaScript, so finance can't update it; the duplicate check only knows one historic invoice, so the same invoice submitted twice is not caught ("Duplicate not detected on second submission"); it treats each invoice alone, so it misses patterns across invoices; every outcome sends the same Telegram message; and the Gemini note in D is never saved. These are exactly the gaps that let duplicate payments and approval-limit splitting through in real accounts-payable teams.

4.2 Design

Workflow	Role
sub | Core rules + AP ledger	Loads the PO master and the whole ledger from Data Tables, runs the V4 rules, writes each case to `Week10_AP_Ledger_V4`. Deterministic; no AI.
sub | Alerts + Gemini explanation	Switch by priority. P1 HOLD: instant Telegram, facts only. P2 REVIEW: Gemini explains, a guardrail checks it, the note is written back to the ledger row, then Telegram. P3 READY: no message (avoids alert fatigue).
E1 AP intake form	Form for AP clerks; dropdown for the bank-change question; shows the triage result and case id when submitted.
E2 Batch invoice API	`POST /webhook/meridian-ap-v4/batch` takes 1–200 invoices (e.g. from an OCR or email-capture system) and returns a JSON summary.
E3 Daily AP exception digest	Every weekday at 08:30 SGT: counts and amounts from the ledger; Gemini writes a 3-bullet briefing; a guardrail checks that every number in it exists in the data.



Core sub-workflow on the batch run: 8 invoices → PO master (7 rows) → ledger → rules → saved.



Alerts sub-workflow: 2 instant HOLD alerts, 4 Gemini-explained reviews (guardrail → ledger update → Telegram), 2 logged only.

4.3 Rules added (precedence order)

#	Status	Condition	New in V4?
1	HOLD_INVALID_DATA	missing field or invalid amount	
2	HOLD_DUPLICATE	same supplier + reference anywhere in the ledger, earlier in the batch, or in the lab fixture (case and spaces ignored)	**persistent**
3	HOLD_BANK_CHANGE	supplier says bank details changed	
4	REVIEW_MISSING_PO	no PO	
5	REVIEW_UNKNOWN_PO	PO not in the PO master **Data Table**	table, not code
6	REVIEW_PO_CLOSED	PO status is not OPEN	**new**
7	REVIEW_PO_SUPPLIER	PO belongs to another supplier	
8	REVIEW_PO_ALREADY_INVOICED	the PO was already used by an earlier invoice with a different reference (re-billing)	**new**
9	REVIEW_AMOUNT_MISMATCH	amount differs from the PO	
10	REVIEW_SPLIT_SUSPECTED	same supplier, last 7 days, each ≤ $10k but together > $10k (approval-limit splitting); also sets manager approval	**new**
11	REVIEW_MANAGER	amount > $10,000	
12	READY_FOR_HUMAN_CHECK	all controls pass	

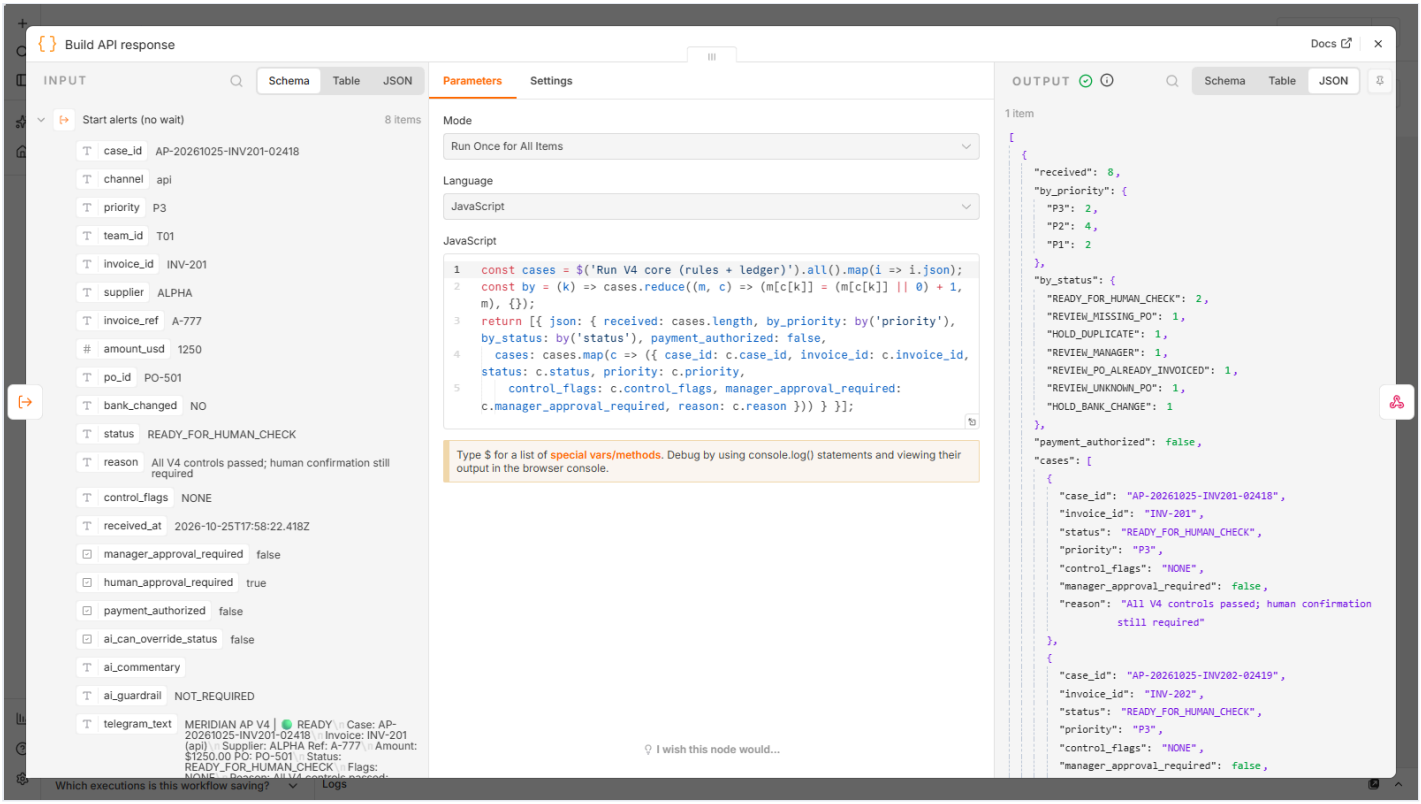
Each case also gets a `case_id` and a priority (P1/P2/P3). `human_approval_required=true`, `payment_authorized=false` and `ai_can_override_status=false` on every row, as in the lab. The rules are a plain function (`tests/classify_v4.js`) that the build script copies into the Code node, with 12 unit tests, including the guide's Appendix A answers for all 8 lab invoices.

5. V4 results

5.1 The lab's 8 invoices sent as one API batch (executions #12, #13, #14)

Invoice	Priority	Status	Flags	Route	AI guardrail
INV-201	P3	READY_FOR_HUMAN_CHECK	NONE	logged, digest only	-
INV-202	P3	READY_FOR_HUMAN_CHECK	NONE	logged, digest only	-
INV-203	P2	REVIEW_MISSING_PO	MISSING_PO|HIGH_VALUE	Gemini note + alert	PASSED
INV-204	P1	HOLD_DUPLICATE	DUPLICATE	instant HOLD alert	-
INV-205	P2	REVIEW_MANAGER	HIGH_VALUE	Gemini note + alert	PASSED
INV-206	P2	REVIEW_PO_ALREADY_INVOICED	PO_ALREADY_INVOICED|SPLIT_SUSPECTED	Gemini note + alert	PASSED
INV-207	P2	REVIEW_UNKNOWN_PO	UNKNOWN_PO	Gemini note + alert	PASSED
INV-208	P1	HOLD_BANK_CHANGE	BANK_CHANGE	instant HOLD alert	-

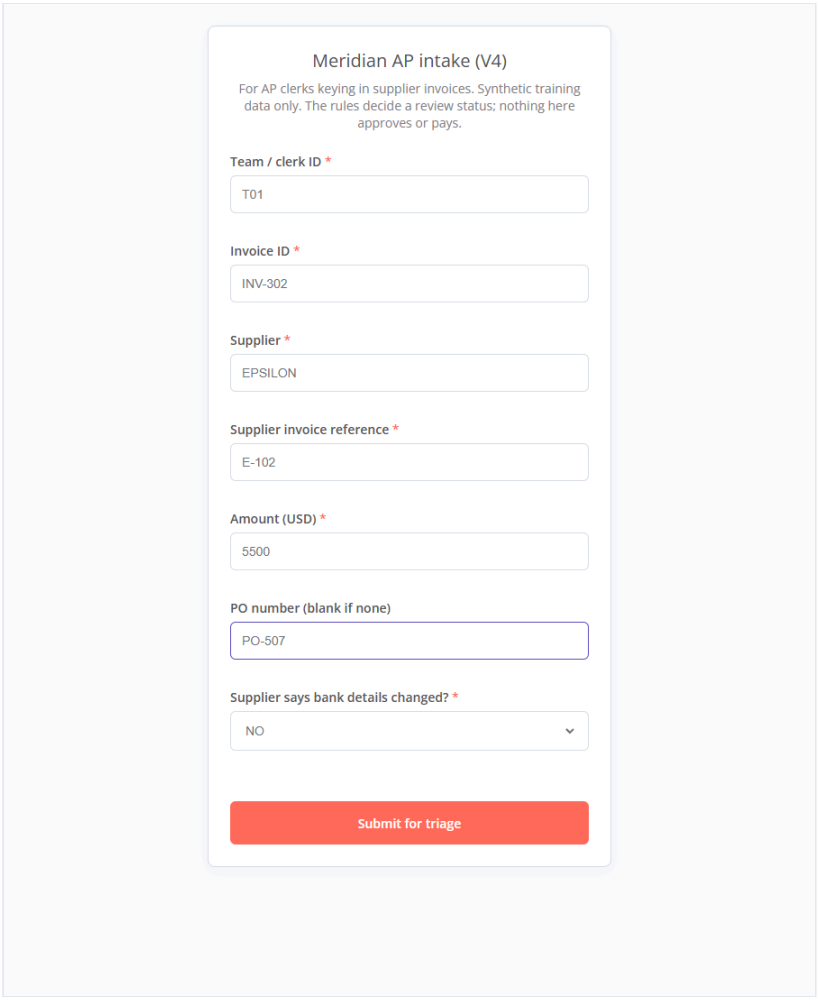
Same answers as the lab for 7 of 8. The difference is **INV-206**: V3 only saw an amount mismatch, while V4 sees that INV-202 already used PO-502 in the same batch (possible re-billing) and that the two BETA invoices ($8,950 + $9,150) look like a split.



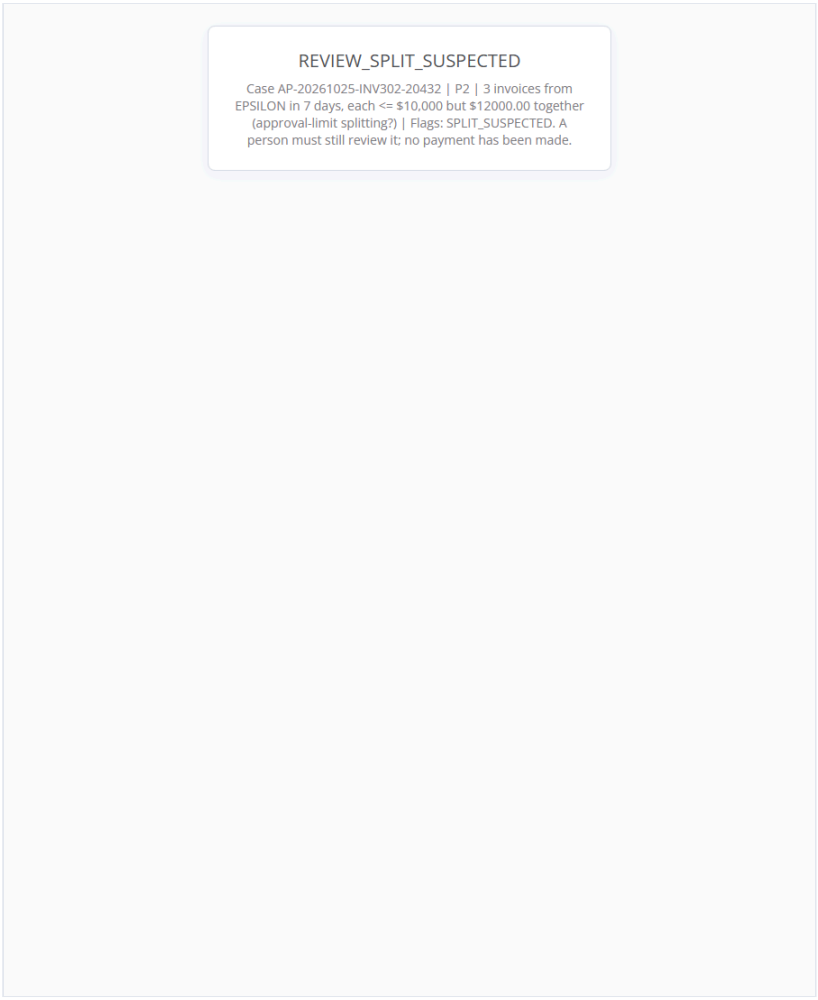
E2: JSON returned to the calling system.

5.2 Scenarios for the new controls

Submitted	Executions (core / alerts)	Status	Reason	Telegram	AI
INV-301 EPSILON $6,000 PO-506 (form)	#16 / #17	READY_FOR_HUMAN_CHECK	All V4 controls passed; human confirmation still required	none (P3)	-
INV-302 EPSILON $5,500 PO-507 (form)	#19 / #20	REVIEW_SPLIT_SUSPECTED	3 invoices from EPSILON in 7 days, each <= $10,000 but $12000.00 together (approval-limit splitting?)	ok=true, message_id 47	PASSED
INV-201B 'alpha' ' a-777 ' $1,250 (form, re-keyed)	#22 / #23	HOLD_DUPLICATE	Supplier and reference already submitted (found in AP ledger table or earlier in this batch)	ok=true, message_id 48	-
INV-303 ZETA $3,200 PO-508 (API)	#25 / #26	REVIEW_PO_CLOSED	PO-508 is CLOSED, not OPEN	ok=true, message_id 49	PASSED



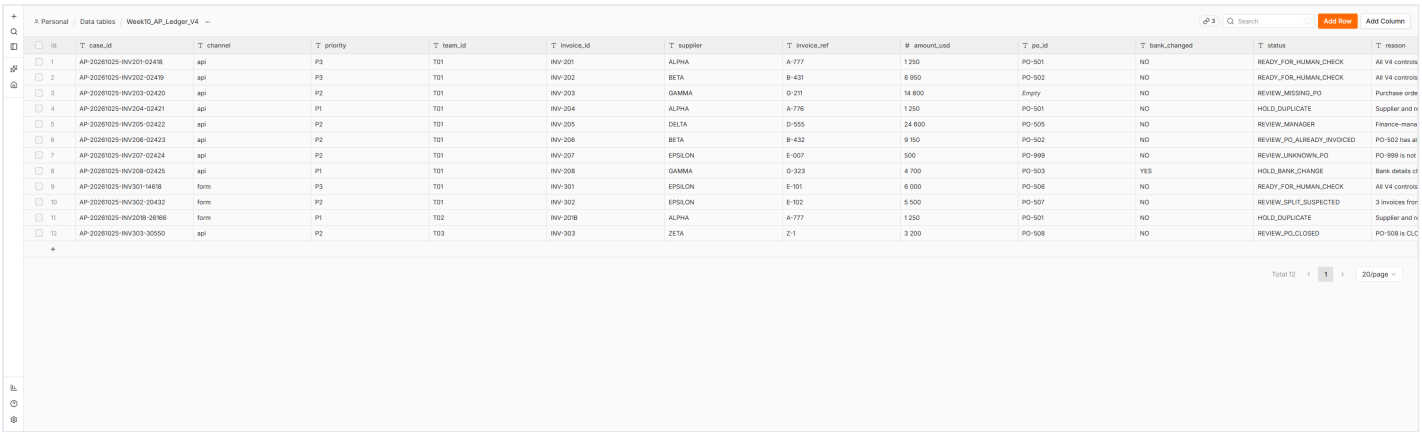
E1: the AP clerk form (INV-302).



E1: result shown to the clerk: split suspected. The rule counted three EPSILON invoices this week (INV-207 $500 from the batch, INV-301, INV-302).

Gemini's note for INV-302 (guardrail PASSED):

```
This invoice presents a risk of invoice splitting to bypass standard approval thresholds, as multiple submissions from EPSILON totaling $12,000.00 were received within a 7-day window. Please verify the business justification for these separate purchases and confirm whether they should have been consolidated under a single authorization.
```



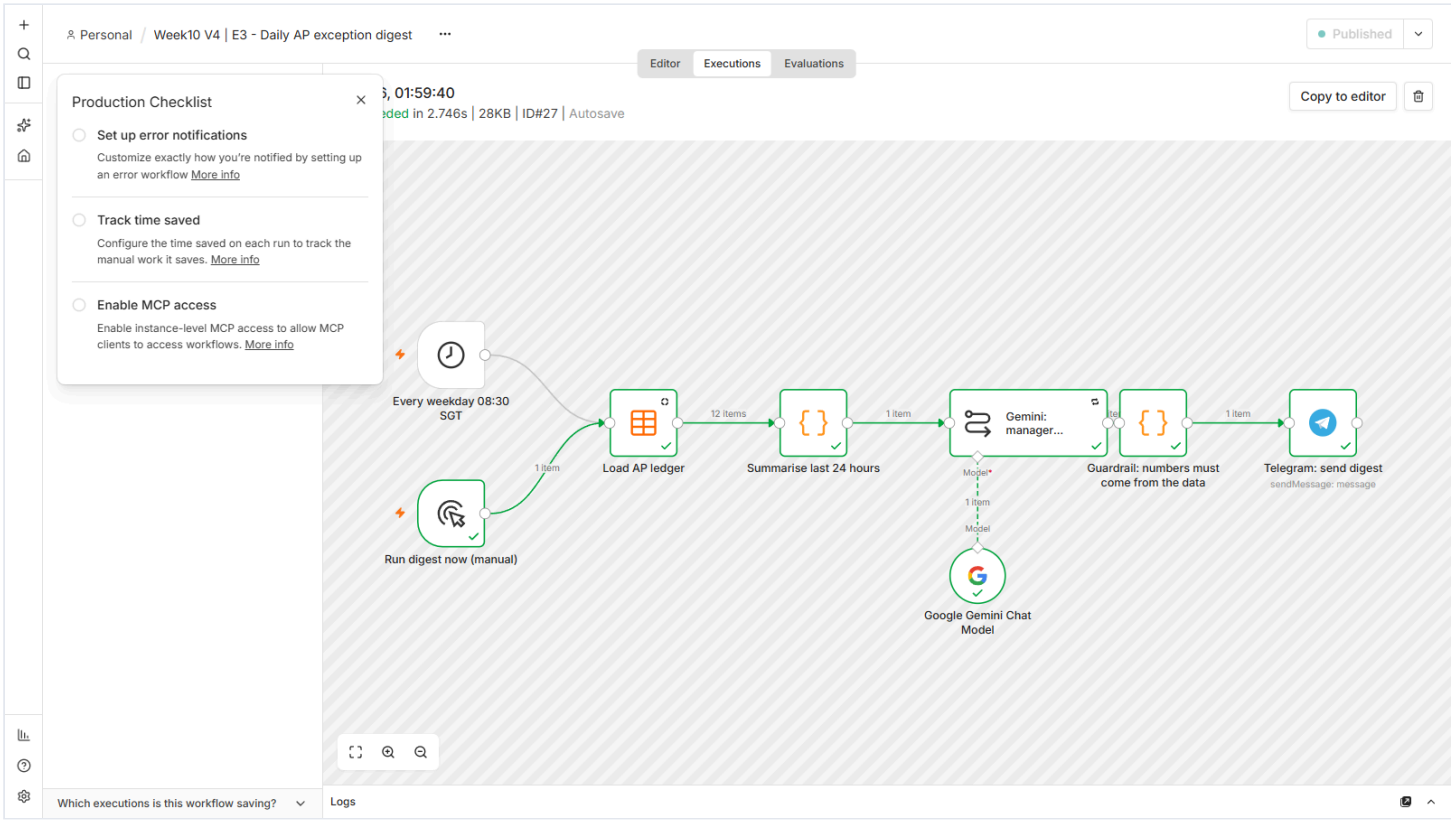
Week10_AP_Ledger_V4 after all scenarios: 12 cases from the API and the form.

5.3 Daily digest (E3, execution #27, guardrail PASSED)

```
MERIDIAN AP V4 | DAILY DIGEST
Cases: 12 | P1 3 / P2 6 / P3 3
On hold: $7200.00 | Needs manager: $54050.00
By status: READY_FOR_HUMAN_CHECK 3, REVIEW_MISSING_PO 1, HOLD_DUPLICATE 2, REVIEW_MANAGER 1, REVIEW_PO_ALREADY_INVOICED 1,
REVIEW_UNKNOWN_PO 1, HOLD_BANK_CHANGE 1, REVIEW_SPLIT_SUSPECTED 1, REVIEW_PO_CLOSED 1

* **Action First:** Review the 1 manager review case worth $54,050.
* **Total Value on Hold:** $7,200.
* **Unusual:** There are 2 duplicate holds for the exact same amount ($1,250 for INV-204 ALPHA and INV-201B ALPHA).

(Advisory summary. Statuses come from the rules; nothing is approved or paid.)
```

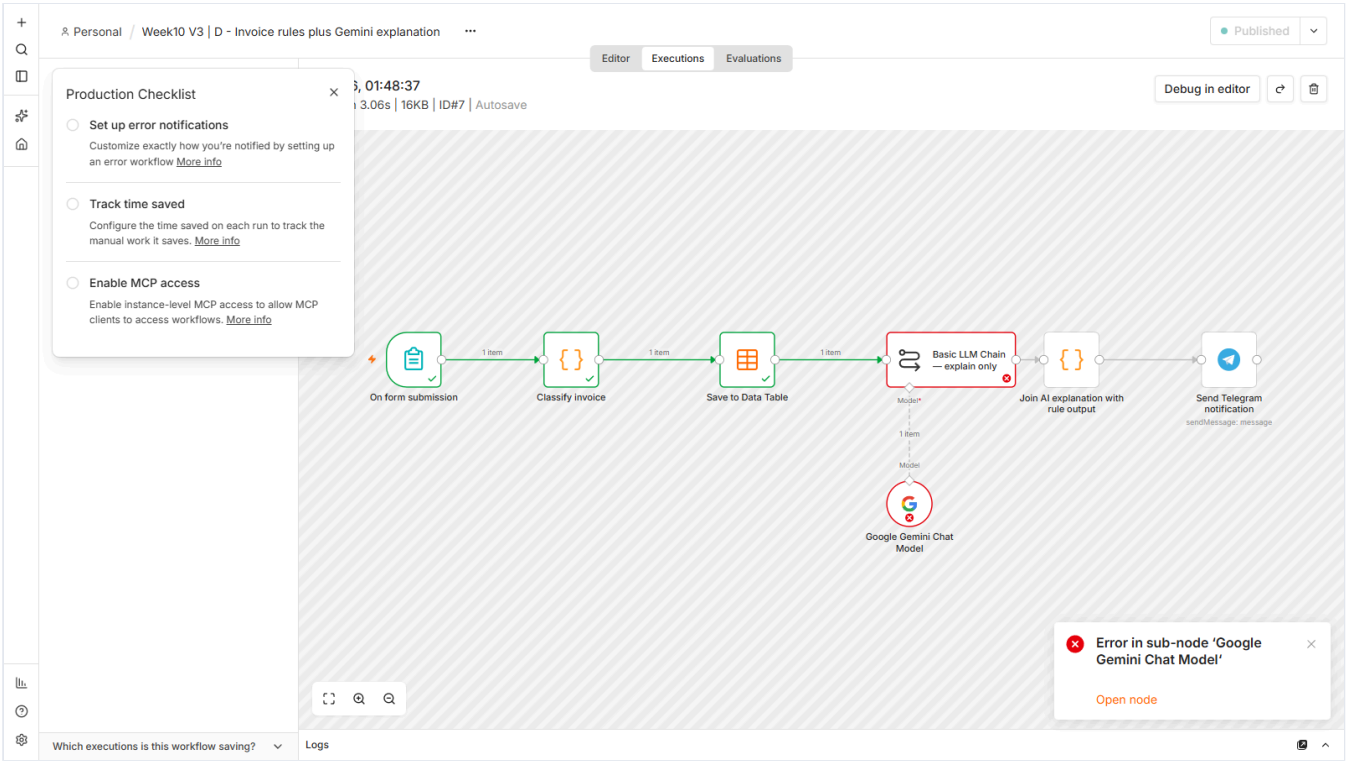


E3: ledger → deterministic summary → Gemini briefing → number guardrail → Telegram.

6. Problems met and how they were fixed

6.1 Gemini 503 (lab D, execution #7)

The template's `models/gemini-3.8-flash` answered "Service unavailable". A direct API check showed 3.8-flash and flash-latest overloaded, while `gemini-flash-lite-latest` answered normally. Following the guide's troubleshooting advice ("choose another currently accessible model"), workflow D now uses flash-lite with n8n's *Retry on fail* (3 tries, 5 s apart). Re-run #8: success.



The failed run, kept as evidence.

6.2 Telegram "can't parse entities" (V4 alerts, first run)

The V4 alert text has status codes like `HOLD_DUPLICATE`, and Telegram read the underscores as Markdown. The lab template avoids this with `parse_mode: None`, which I had left out of my new Telegram nodes. Fixed, ledger reset, all scenarios re-run.

6.3 Other setup issues

- n8n 2.x needs Node ≥ 24, so I used a portable Node 24 rather than changing the global install.
- n8n 2.x will only publish a workflow if the sub-workflows it calls are published, so the setup script publishes sub-workflows first.

7. AI boundaries and reflection

Is this an agent? No. As the guide says, these are fixed workflows with a model in an advisory slot. In V4 the model writes text only; it cannot change a status, call a tool, approve or pay. Its output passes through code guardrails, tested in `tests/test_guardrails.js` against claims like "has been approved for payment", requests for bank details, a different status name, and invented numbers in the digest.

Two places where deterministic rules beat LLM judgement:

1. **Duplicate and re-billing detection.** Whether supplier + reference already exists, or a PO is already used, is an exact lookup over every earlier invoice. Rules get it right every time and can be audited; a model reading invoices one at a time cannot see the history and could be talked out of it.
2. **Approval thresholds and split detection.** "> $10,000" and "the sum this week > $10,000" are policy, not opinion. They must give the same answer on every run, and an auditor must be able to replay them. The model only explains why the rule fired.

Limitations:

- **Static fixtures (lab):** V3's PO master and one-row history are constants in code, so they go stale and miss real duplicates. V4 moves both into Data Tables, but the PO table is still a manual copy, not a live ERP feed.

- **Guardrails check form, not meaning.** In the digest, Gemini wrote "the 1 manager review case worth $54,050", but $54,050 is the total across four cases needing a manager. Each number exists in the data, so the guardrail passed it. The deterministic header above the AI text is the figure to rely on.
- Split detection uses supplier name and a 7-day window; a real system would also match supplier ids, bank accounts and invoice dates.
- The local n8n is not reachable from the internet (forms and the webhook work on this machine only), and the webhook has no authentication. Production would need HTTPS, header auth and a unique constraint on supplier + reference.

8. Files

Path	Contents
workflows/lab_*.json	Lab workflows A–D as configured (chat id as an env expression; D on flash-lite with retries)
workflows/v4_*.json	The five V4 workflows (generated by tools/build_v4_workflows.js)
execution_logs/	Every execution: per-node status and full data, decoded from n8n's format, personal identifiers masked
screenshots/	Forms, canvases with run data, node outputs, Data Tables, workflow and credential lists
data/	Table schemas, PO master seed, API test requests and responses, CSV exports of all four Data Tables
tests/	classify_v4.js (rules) + 12 rule tests + 10 guardrail tests (node tests/test_*.js)
tools/	n8n_setup.py (rebuild), run_v4_scenarios.py, submit_form.py, capture_screenshots.py, build_v4_workflows.js